

Which Regression Line Fits?

AP Statistics · Unit 5 · Topic 5.5 · B: 2027-02-12 · E: 2027-04-09 · TEACHER KEY

CED TETHER

- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU DAT-1** — Regression models may allow us to predict responses to changes in an explanatory variable.
- **LO DAT-1.G** — Estimate parameters for the least-squares regression line model. [Skill 2.C]
- **EK DAT-1.G.1** — The least-squares regression model minimizes the sum of the squares of the residuals and contains the point $(\bar{x}, \bar{y})$.
- **EK DAT-1.G.2** — The slope, b , of the regression line can be calculated as $b = r(s_y / s_x)$, where r is the correlation between x and y , s_y is the sample standard deviation of the response variable, and s_x is the sample standard deviation of the explanatory variable.
- **EK DAT-1.G.3** — Sometimes, the y -intercept of the line does not have a logical interpretation in context.
- **EK DAT-1.G.4** — In simple linear regression, r^2 is the square of the correlation, r . It is also called the coefficient of determination. r^2 is the proportion of variation in the response variable that is explained by the explanatory variable in the model.
- **LO DAT-1.H** — Interpret coefficients for the least-squares regression line model. [Skill 4.B]
- **EK DAT-1.H.1** — The coefficients of the least-squares regression model are the estimated slope and y -intercept.
- **EK DAT-1.H.2** — The slope is the amount that the predicted y -value changes for every unit increase in x .
- **EK DAT-1.H.3** — The y -intercept value is the predicted value of the response variable when the explanatory variable is equal to 0. The formula for the y -intercept, a , is $a = \bar{y} - b\bar{x}$.

Essential question: How do several regression properties distinguish lines that share the same mean point?

DOK-3 skill: 4.B · **FRQ pattern:** shared-mean-point-which-line-is-lsrl

DOK rationale: Part (c) synthesizes a warranted regression analysis from two partly defensible accounts by coordinating slope units, least-squares and mean-point properties, coefficient meanings, domain, and prediction consequences.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Ask students to mark one defensible and one questionable clause before choosing a line.
- After the video, require slope units, the mean-point property, and r^2 evidence.
- Collect at the door; score part (c) only, E/P/I.

Students do

- Make a first claim, then complete the 2.8 follow-along.
- Synthesize a supported analysis in (c), revise, and turn in.

Questions to ask

- Which standard-deviation ratio has response-per-explanatory units?
- Why can both lines contain (7, 28)?
- What differs at 8 sunshine hours?

Adult role

Circulate. Ask for linked evidence rather than confirming an analyst.

[WARN] WATCH FOR

- Reversing s_y/s_x .
- Treating the mean point as sufficient.
- Using r rather than r^2 for explained variation.

SCORING PART (c) — E / P / I

Expected elements

- **selects-supported-components** (required) — Retains Quinn's slope and line plus Rowan's r^2 and domain caution
- **coordinates-line-properties** (required) — Uses slope units and explains why the mean point is necessary but insufficient
- **interprets-coefficients** (required) — Interprets slope, intercept, and 64 percent with the observed domain
- **quantifies-consequences** (required) — Gives 31.2 versus 28.2 MWh and the 16-point explained-variation error

E	Synthesizes the correct line and interpretations, links the regression properties, and quantifies both consequences.
P	Finds the correct line and r^2 but incompletely links a property, domain, or consequence.
I	Accepts the reversed ratio, treats the mean point as sufficient, or reports 80 percent explained.

Common mistakes

- Reversing the standard-deviation ratio
- Treating the mean point as sufficient
- Using r as explained variation

[STAR] TODAY'S PROBLEM — annotated

Suppose a solar-farm analyst uses hours of direct sunshine, x , to predict daily generation, y , in megawatt-hours. Summary statistics for 12 observed days are shown.

Quinn: “Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through $(7, 28)$, which by itself proves it is the LSRL. Since $r = 0.80$, the model explains 80 percent of the variation.”

Rowan: “Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through $(7, 28)$, so that line is equally eligible. Explained variation is 64 percent, and a zero-hour intercept needs caution because 0 is outside the observed range.”

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE (5 min)

Which clauses in Quinn’s and Rowan’s analyses look defensible? Cite one formula or line property.

Key: Not graded. Credit a committed clause-level comparison supported by one property.

Rules callout “MAKE THE REGRESSION PROPERTIES AGREE (keep on the page)” is printed on the student sheet.

DOK 1 (a) State what the LSRL minimizes, the point it contains, and the contextual meanings of slope and intercept.

Key: The LSRL minimizes the sum of squared residuals and contains $(\bar{x}, \bar{y})$. Its slope is the predicted response change for a one-unit increase in x ; its intercept predicts the response at $x = 0$.

DOK 2 (b) Calculate the LSRL slope and intercept, r^2 , and the prediction at 8 sunshine hours. Verify the mean-point property.

Key: $b = 0.80(6/1.5) = 3.2$ MWh per hour and $a = 28 - 3.2(7) = 5.6$ MWh, so $\hat{y} = 5.6 + 3.2x$. At $x = 7$, $\hat{y} = 28$; $r^2 = 0.64$; and $\hat{y}(8) = 31.2$ MWh.

DOK 3 (c) Evaluate both analyses and write one warranted synthesis. Coordinate the slope ratio and units, least-squares and mean-point properties, coefficient meanings, and the observed domain. Compare both lines at 8 hours and quantify the consequences of using the rejected slope and using r instead of r^2 .

Key: Quinn has the correct line because s_y/s_x has response-per-explanatory units, but the shared point alone is not sufficient and explained variation is not r . Rowan correctly squares r and recognizes the domain concern, but reverses the slope ratio. Both lines contain $(7, 28)$ because their intercepts were set by $a = \bar{y} - b\bar{x}$; only the 3.2-slope line has the regression slope and minimizes squared residuals. The supported line predicts 31.2 MWh at 8 hours; the other predicts $26.6 + 0.2(8) = 28.2$ MWh, 3.0 MWh lower. Explained variation is 64 percent, so using 80 percent overstates it by 16 percentage points. The correct 5.6-MWh intercept is a prediction at 0 hours, outside the observed 4–10-hour range, so it is unsupported here.

EXIT

One thing the video changed about my first take: _____